High-Speed Internet and Government Responsiveness: Evidence from U.S. Federal Spending

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Abstract

This paper studies how broadband Internet diffusion affects government responsiveness by examining the allocation of U.S. federal spending across municipalities between 1999 and 2010. Using proximity to the Interstate Highway System as a cost-based instrument for broadband deployment, I find that broadband expansion increases discretionary spending, the component most responsive to political incentives, while leaving mandatory spending unaffected. Broadband diffusion also increases campaign contributions in presidential and congressional elections, partly driven by the entry of new donors. These findings suggest that broadband diffusion strengthens electoral incentives and shifts public spending toward domains where elected officials retain allocation discretion.

Keywords: Broadband Internet, Electoral Incentives, Government Spending, Political Engagement, Campaign Contributions

JEL Codes: D72, H40, H50, L86, O33

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1 Introduction

Traditionally, mass media has played a central role in the political process by providing voters with information that enables them to monitor and discipline politicians' behavior. As documented in theoretical and empirical work, greater voter information through offline media makes political decisions more sensitive to government performance, thereby strengthening electoral incentives and inducing politicians to be more responsive (Besley and Burgess, 2002; Strömberg, 2004, 2015; Ferraz and Finan, 2008). In the context of digital technologies, recent work shows that these technologies can also enhance political accountability and increase electoral competition by expanding access to information and lowering barriers to political participation (Guriev et al., 2021, 2025; Petrova et al., 2021; Enríquez et al., 2024). However, how these changes translate into greater government responsiveness remains an open question.

This paper studies how broadband Internet diffusion affects government responsiveness by examining the allocation of U.S. federal spending across local governments at the subcounty level (hereafter, municipalities) between 1999 and 2010, the period of fastest broadband diffusion. I focus on how broadband Internet expansion affects the composition of spending across categories that differ in their exposure to political discretion. To empirically assess this relationship, I combine data on federal government expenditure from the Consolidated Federal Funds Report (CFFR) with information on broadband Internet services from Form 477 collected by the Federal Communications Commission (FCC). Because granular data on broadband subscriptions are not available for most of the period, I use the number of Internet service providers (ISPs) serving a municipality as a proxy for broadband diffusion. I show that ISP entry is positively associated with subscription growth during this period, reflecting the role of local competition in expanding access.

Identifying the causal effect of broadband diffusion is challenging, as broadband deployment may respond to differences in local economic conditions and political influence, leading to omitted variable bias and reverse causality. To address these concerns, I develop a cost-based identification strategy that exploits geographic variation in broadband deployment costs. Specifically, I use municipalities' proximity to the Interstate Highway System as an instrument for broadband diffusion. Fiber-optic backbone infrastructure in the United States is typically installed along Interstate rights-of-way (Durairajan et al., 2015), and hence municipalities located closer to these corridors face lower connection costs. Because the Interstate Highway System was largely completed decades before the emergence of commercial broadband, its geographic layout provides plausibly exogenous variation in broadband deployment costs. I exploit this source of variation in an instrumental-variable framework that allows the effect of proximity to evolve over time with the diffusion of broadband.

Using this instrumental-variable approach, I document that broadband expansion does not lead to a statistically significant increase in total federal transfers, but instead changes their composition. In particular, broadband diffusion significantly increases discretionary federal transfers, while having no effect on mandatory, formula-based programs. This distinction is economically and institutionally meaningful. While mandatory spending is governed by statutory entitlement rules and is largely insulated from short-run political pressures, discretionary spending is determined annually through the appropriations process and is more susceptible to electoral incentives. The increase in discretionary spending is broad-based and not driven by any single program category, pointing to a general shift in how resources are allocated rather than to changes in specific policies.

This pattern is difficult to reconcile with a purely mechanical increase in program take-up, which would predict effects on mandatory transfers. Instead, the concentration of effects in domains where policymakers retain discretion is consistent with the broader interpretation that, by expanding access to political information and lowering barriers to political engagement, broadband Internet may sharpen electoral incentives by making citizens more likely to hold politicians accountable.

To assess the plausibility of this interpretation, I next examine the effect of broadband on a concrete and policy-relevant form of political participation: campaign contributions. Using administrative campaign finance records from the Database on Ideology, Money in Politics, and Elections (Bonica, 2024), I show that broadband expansion increases campaign contributions in presidential and congressional elections. These effects operate along the intensive margin and, importantly, along the extensive margin, through the entry of new donors and a rise in small contributions. These findings indicate that broadband diffusion broadened participation in campaign finance, expanding the set of individuals who actively engage in politics. While this analysis does not establish that contributions mediate the effect of broadband on federal spending, it shows that broadband also increases a form of political engagement that may strengthen citizens' ability to hold politicians accountable.

I conduct a series of robustness and falsification tests to support the validity of the identification strategy. First, I find no evidence of differential pre-trends in federal spending prior to broadband diffusion. Second, I show that the estimates are not driven by potentially confounding shocks or economic factors that may be correlated with municipalities' proximity to the Interstate

Highway System. In particular, proximity does not predict differential growth in discretionary defense outlays following 9/11 and the onset of the *war on terror*, and the estimates are robust to flexibly controlling for the location of military installations. The findings are also not explained by changes in local economic activity or the location of freight and distribution facilities. Third, proximity does not predict changes in transportation-related federal spending. Finally, I show that the findings are not driven by the political considerations that shaped the original layout of the Interstate Highway System.

This paper contributes to three strands of the literature. First, it contributes to the literature on media and political accountability. A long tradition of theoretical and empirical work establishes that in models of retrospective voting with imperfect information, media can enhance government responsiveness by informing voters on government performance, which they use to guide their political decisions (Besley and Burgess, 2002; Strömberg, 2004; Snyder and Strömberg, 2010; Strömberg, 2015). Recent work extends these insights to the digital environment, showing that broadband Internet access and social media can improve political accountability and intensify electoral competition by expanding access to information and lowering barriers to political participation (Guriev et al., 2021, 2025; Petrova et al., 2021; Enríquez et al., 2024). This paper contributes to this literature by providing causal evidence on how these changes translate into government behavior.

Second, this paper contributes to the literature documenting the political effects of the broadband Internet. While existing work shows that digital technologies affect political behavior and accountability (Zhuravskaya et al., 2020), less is known about how these changes translate into government decisions over the allocation of public resources. My findings also help reconcile contrasting evidence across contexts by highlighting the role of underlying mechanisms. In particular, Gavazza et al. (2019) show that broadband diffusion in the United Kingdom is associated with lower local government spending, which they interpret as a shift toward entertainment consumption and away from political engagement. In contrast, I show that in settings with greater information frictions, such as the U.S. federal context, where spending is mediated by complex legislative and appropriations processes, broadband Internet access can instead strengthen political engagement and responsiveness.

My findings also differ from Bessone et al. (2022), who show that mobile Internet and social media in Brazil lead politicians to substitute offline transfers with online engagement. In my setting, broadband diffusion largely predates the widespread use of social media as a substitute for offline political activity. Consequently, I argue that the relevant margin is not the substitution of offline transfers with online engagement, but rather the strengthening of electoral incentives through improved voter information and participation, which in turn affects policy choices.

Finally, the paper contributes to the literature on campaign finance. A central concern in this literature is that political giving is highly concentrated among a small share of the population, raising questions about whose preferences are reflected in policy (Cagé, 2024). Recent work shows that digital technologies help candidates expand their donor bases by allowing them to reach donors more effectively (Petrova et al., 2021). Because both their identifying variation and outcomes are defined at the candidate level, a larger donor base may reflect either the entry of new participants into campaign finance or the reallocation of existing donors toward technology-adopting candidates. By identifying first-time contributors across candidates at the municipality level, I isolate the former and show that broadband diffusion expands the overall donor pool. These effects are consistent with the emergence of online fundraising platforms such as ActBlue, which lowered the transaction costs of political giving and expanded participation beyond the traditionally narrow donor base.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background on federal spending and broadband expansion. Section 3 presents the data, and Section 4 outlines the empirical strategy. Section 5 reports the main results. Section 6 examines the effects of broadband on campaign contributions. Section 7 concludes.

2 Institutional Setting

2.1 Broadband Internet as a Political Tool in the U.S.

Political use of the Internet in the United States began in the early 1990s with the spread of dial-up connections. During this period, the Internet primarily complemented traditional media, enabling political actors to disseminate information, communicate with politically engaged audiences, and raise campaign funds at relatively low cost. For the mass electorate, however, political use remained limited. Adoption rates were modest, email was the dominant application (Bimber and Davis, 2003; NTIA and RUS, 2000), and most online news content closely mirrored print sources (Li, 2013). As a result, dial-up Internet did not substantially alter the information environment faced by most voters. The diffusion of broadband Internet represented a qualitatively different shift. Broadband enabled continuous, high-speed connections and sharply reduced the cost of accessing information, expanding both the volume and timeliness of political content available online. This transition coincided with a rapid increase in political uses of the Internet: the share of Internet users obtaining political news online rose from 33% in 2000 to

66% in 2008, and by the 2008 election cycle nearly half reported engaging in political activity online (Smith, 2009). Although adoption was uneven across locations, broadband expanded rapidly during the 2000s and reshaped how political information was accessed and shared.

During the period of analysis, broadband expansion overlapped with the emergence of early social media platforms, including MySpace (2003), Facebook (2004), YouTube (2005), and Twitter (2006). In the 2000s, these platforms were still in relatively early stages and operated alongside blogs, online forums, and other forms of user-generated content, rather than dominating political communication as later social media would (e.g., during the 2008 presidential election, only 10% of U.S. adults reported using social networking sites to engage in political activity (Smith and Rainie, 2008)). Together, these technologies reflected a broader shift toward more interactive and participatory forms of political communication enabled by broadband access. By lowering barriers not only to accessing information but also to producing and disseminating content, broadband plausibly strengthened voters' ability to monitor political performance, share information within communities, and coordinate political activity.

At the same time, improved broadband access may have affected political responsiveness through institutional channels beyond voter behavior. In the U.S., reducing informational and administrative frictions through Internet-based technologies has been an explicit policy objective since the early 2000s. The *E-Government Act of 2002* and subsequent initiatives by the *Office of E-Government and Information Technology* within the Office of Management and Budget are examples that explicitly promoted the use of Internet-based technologies to improve access to government information and reduce communication costs. In this context, broadband expansion may have increased the capacity of local governments and citizens to learn about programs, prepare applications, and navigate federal funding processes, potentially affecting the geographic distribution of federal spending.

2.2 U.S. Federal Spending

The allocation of federal spending across municipalities provides a useful setting for examining whether changes in the media environment affect government responsiveness. First, federal spending consists of two broad categories, mandatory and discretionary, whose levels and composition are determined through the annual budget process involving both the President and Congress. Mandatory spending accounts for roughly two thirds of total federal expenditures and includes entitlement programs such as Social Security and Medicare, as well as interest payments on the public debt. These programs are authorized by permanent legislation and governed by statutory eligibility rules, leaving relatively limited scope for short-run political discretion or targeted geographic reallocation (Levit et al., 2015).

By contrast, discretionary spending is authorized through the annual appropriations process and accounts for approximately one third of the federal budget. It includes national defense, public infrastructure investment, and a wide range of grant and procurement programs administered across different levels of government (Austin, 2014). Because discretionary funding is set and renewed annually through appropriations bills, Congress and the President exercise substantial influence over program levels and their geographic distribution.[1] As a result, discretionary spending is generally more sensitive to political incentives than mandatory programs.

Second, a substantial share of discretionary federal spending is geographically allocated and tied to specific localities. While funds often flow through grants or procurement contracts administered by federal or state agencies, their economic incidence is local, supporting activities and beneficiaries within particular municipalities. Programs related to disaster assistance, housing, transportation, and education operate at relatively fine geographic levels and are often visible to local residents (U.S. Census Bureau, 2018). This geographic targeting exposes municipalities, and the populations residing within them, to meaningful variation in discretionary federal spending over time.

3 Main Datasets and Descriptive Statistics

This section describes the main data sources used in the empirical analysis. I combine municipality-level data on federal spending with geographically granular measures of broadband Internet services, supplemented by sociodemographic, economic, and geographic controls. Because variables are reported at different geographic units, I aggregate all data to the municipality level using 2002 Census of Government boundaries.[2] The resulting dataset is a panel of 35,672 municipalities observed annually from 1999 to 2010.

3.1 Federal Expenditure

I use data on federal outlays and obligations to state and local governments from the Consolidated Federal Funds Report (CFFR) for the period 1983–2010. The dataset records annual federal transfers by recipient jurisdiction and program, covering more than 3,000 distinct federal

[1] Continuing resolutions, which temporarily extend prior-year funding levels when new appropriations are delayed, constrain new initiatives in the short run but do not eliminate discretion over the geographic implementation of existing programs (Congressional Research Service, 2019).

[2] I use the term *municipality* to refer to general-purpose subcounty local governments, corresponding to the Census geographies of Incorporated Places and Minor Civil Divisions.

programs. Each program is classified into broad functions and object categories using definitions consistent with the federal budget. I use the object classification to distinguish between mandatory spending, discretionary spending, and loans and insurance programs within each budget function. Loans and insurance programs generally do not involve immediate cash outlays; rather, they capture federal financial guarantees and risk-sharing arrangements that compensate beneficiaries or cover losses when adverse events occur.

For most transfers, the CFFR reports the place of residence of the recipient, while for procurement contracts it records the place of performance, which corresponds to the location where the contracted activity takes place. These locations may correspond to states, counties, or municipalities. The empirical analysis focuses on federal spending that can be geographically assigned and consistently linked to municipal jurisdictions over time. Studying spending at the municipality level, rather than the county level, exploits the finest geographic level at which these transfers are recorded. It also preserves variation in broadband access across municipalities within the same county, which county-level analysis would average away. Although this measure does not capture all federal expenditures that may ultimately benefit residents of a municipality, it reflects allocations for which geographic targeting is explicit in administrative records. As a result, it captures a component of federal spending that can plausibly vary across locations in response to local political conditions.

Table 1 shows that municipalities receive on average $2,688 per capita in federal spending over the sample period. Approximately 40% of this amount corresponds to discretionary spending, which is determined annually through the congressional appropriations process. Most of the remaining transfers are associated with loans and insurance programs, while a smaller share corresponds to mandatory payments governed by statutory entitlement rules.

Table 1: Summary Statistics for Main Variables, 1999–2010

	Mean	SD	Min	Max
Federal Spending per Capita				
Total	2,687.489	7,689.715	0	1,972,650
Mandatory	187.802	560.079	0	55,975
Discretionary	1,064.474	3,330.189	0	1,971,836
Loans and Insurance	1,435.213	6,741.279	0	1,103,285
Broadband Services				
Broadband Availability	0.990	0.101	0	1
Number of ISPs	8.879	4.777	0	35

Notes: This table reports summary statistics for the main variables used in the analysis over 1999–2010. The unit of observation is municipality-year. Federal spending variables are measured in per capita terms, dividing spending in each municipality by municipality population in 2000. *Broadband Availability* is an indicator variable equal to one if at least one ISP offers service in the municipality in a given year, and *Number of ISPs* is the average number of Internet Service Providers offering service in the municipality. All statistics are weighted by municipality population in 2000.

To further characterize the composition of spending, Appendix Figure A.1 presents the distribution of total federal spending across major budget functions. As shown in the figure, spending is highly concentrated in a small number of categories, with *Community and Regional Development* accounting for approximately 44% of all transfers, followed by *Income Security* (about 8%) and *Education, Training, Employment, and Social Services* (about 8%). While this aggregate distribution highlights the importance of place-based and social policy programs in municipality-level transfers, it masks substantial heterogeneity in the institutional form through which funds are allocated within each function.

To unpack this dimension, Appendix Figure A.2 also reports, for each function, the share of spending accounted for by mandatory, discretionary, and loans and insurance programs. As shown, there is wide variation in composition across functions. For example, *Community and Regional Development* is overwhelmingly composed of loans and insurance programs, reflecting the central role of disaster relief and related compensation mechanisms in this spending domain. In contrast, functions such as *Education, Training, Employment, and Social Services* and parts of *Income Security* exhibit a much larger discretionary component, while programs like *Social Security* are almost entirely mandatory.

Taken together, this composition indicates that a substantial share of the spending captured in the data flows through channels where allocation may be responsive to political incentives.

At the same time, the importance of mandatory and insurance-based programs highlights that not all components of spending are equally subject to such forces. This variation motivates an empirical approach that distinguishes between these margins when assessing the effects of changes in the media environment.

The next subsection describes the data on broadband diffusion used in the analysis.

3.2 Broadband Internet Services

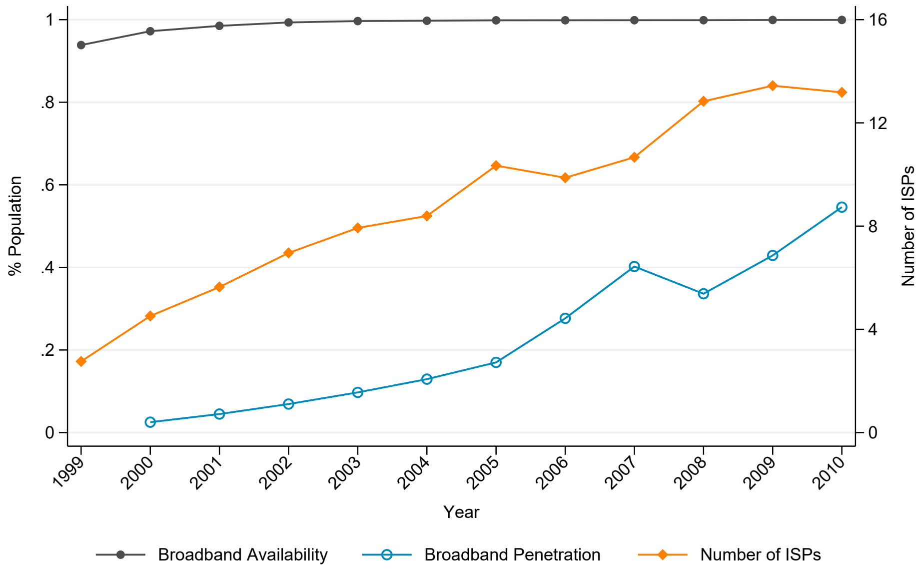
I use data from the Federal Communications Commission (FCC) Form 477, which collects information on the provision of high-speed Internet services by Internet service providers (ISPs). Because detailed data on broadband subscriptions at fine geographic levels are only available starting in 2008, I measure local broadband access using the average number of ISPs offering high-speed Internet service in a municipality.[3] The use of this measure is motivated by descriptive evidence suggesting that broadband adoption depends on service availability and cost (NTIA and RUS, 2000; NTIA, 2007). Appendix B provides supporting evidence that variation in the number of providers is positively correlated with broadband subscriptions, consistent with this measure capturing differences in local broadband availability and the affordability of service. Since this proxy measures broadband adoption with error, the instrumental-variable strategy developed below also serves to attenuate the resulting bias.

Figure 1 illustrates the evolution of the broadband market over the period 1999–2010. Broadband availability was already widespread by 1999, as more than 90% of the population lived in areas served by at least one provider. Despite this broad coverage, broadband adoption remained substantially lower at that time, with less than 5% of the population using broadband at home. Over the following decade, however, local broadband markets expanded rapidly. At the beginning of the period municipalities typically had access to between one and three providers, whereas by 2010 the average municipality was served by roughly thirteen providers. Over the sample period as a whole, municipalities were served on average by nine ISPs, corresponding to roughly one provider per 1,000 residents (Table 1).[4]

[3] Data on Internet service providers are reported at different geographic levels over the sample period. To construct a consistent measure over time, I harmonize FCC data reported at different geographic levels into Census 2000 Zip Code Tabulation Areas (ZCTAs) and aggregate them to municipalities.

[4] Broadband competition in the United States is highly local. Between 1999 and 2010 an average of 58 ISPs operated within a state, with approximately 875 providers nationwide. Growth during this period was driven primarily by the entry of fiber-based providers, while entry in earlier DSL and cable technologies stagnated (FCC, *Internet Access Services Reports*).

Figure 1: Evolution of Broadband Services and Adoption



Notes: This figure shows trends in Broadband availability, adoption, and ISP presence in the United States from 1999 to 2010. *Broadband Availability* is an indicator equal to one if at least one ISP offers service in a municipality in a given year. The population-weighted mean of this indicator across municipalities, shown in the figure, corresponds to the share of the population living in municipalities with at least one ISP (dots, left axis). *Subscriptions* is a measure of broadband adoption, defined as the number of high-speed Internet connections divided by the U.S. population (circles, left axis). *Number of ISPs* is the population-weighted mean number of ISPs offering service in a municipality (diamonds, right axis). Municipality-level measures are weighted by municipality population in 2000.

Despite the rapid expansion of broadband markets, access to broadband services remained uneven across locations. Appendix Table A.1 shows that municipalities with limited or no broadband availability at baseline differed systematically from those with greater ISP presence. In particular, these municipalities were more likely to be rural, less densely populated, have lower income and educational attainment, and be located farther from major metropolitan areas and state capitals. Similar patterns emerge when comparing municipalities with above- and below-median ISP presence among those already covered. These differences are consistent with higher infrastructure deployment costs and weaker expected demand for broadband services. They also highlight the importance of accounting for systematic determinants of broadband deployment when estimating the effects of broadband diffusion.

3.3 Municipality Characteristics

I construct a set of baseline sociodemographic, economic, and geographic characteristics of municipalities using data drawn primarily from the U.S. Decennial Census 2000. These variables include measures of population size and density, age structure, racial composition, educational attainment, veteran status, income, unemployment, industry composition, and rural population share. To capture geographic location and accessibility, I additionally include measures of proximity to the coast, state capitals, and major U.S. cities, as well as elevation. The complete list of municipality characteristics and their summary statistics are reported in Appendix Table A.2.

All monetary amounts are expressed in constant 2000 dollars.

4 Empirical Strategy

To examine whether broadband diffusion affects the allocation of federal spending across U.S. municipalities, I exploit the panel structure of the data and estimate the following specification:

$$\text{Federal Expenditure}_{mt} = \beta \text{Number of ISPs}_{mt} + X'_{mt}\theta + f_m + f_{st} + \varepsilon_{mt} \qquad (1)$$

where the subscript mt refers to municipality m in year t. Federal Expenditure$_{mt}$ is the logarithm of total federal expenditure per capita allocated to a municipality. Number of ISPs$_{mt}$ denotes the average number of high-speed Internet Service Providers operating in the municipality. The vector X_m includes baseline sociodemographic, economic, and geographic characteristics from the 2000 Decennial Census, interacted with year fixed effects.[5] This specification allows municipalities with different initial characteristics to follow differential time trends, while avoiding the inclusion of time-varying covariates that may themselves be affected by broadband Internet diffusion. The model further includes municipality fixed effects, f_m, to absorb time-invariant local characteristics, and state-year fixed effects, f_{st}, which flexibly control for state-level shocks. The error term is denoted by ε_{mt}. All regressions are weighted by municipality population as of 2000, and standard errors are clustered at the state level.[6] The analysis focuses on continental U.S., excluding Alaska, Hawaii, and U.S. outlying territories, and the District of Columbia.

The coefficient of interest is β, which measures how changes in local broadband diffusion, proxied by the number of ISPs, are associated with changes in federal expenditure per capita within a municipality over time. A key challenge in estimating β is that the geographic distribution of ISPs is not random. Internet service providers tend to enter markets with higher expected demand and lower deployment costs, which are correlated with population density, income, and other local characteristics that may independently influence federal spending (Greenstein, 2020).

[5] See Appendix Table A.2 for the full list of controls included in the regression.

[6] Weighting by population ensures that the estimates reflect population-weighted average effects rather than treating municipalities of very different sizes equally, and municipality fixed effects absorb time-invariant differences in size and governance structure across the diverse set of subcounty jurisdictions.

While municipality fixed effects and the flexible inclusion of controls mitigate many sources of confounding, time-varying unobservables related to economic growth, political influence, or administrative capacity may still bias OLS estimates.

A second concern is reverse causality. Although broadband provision in the United States is largely private, federal programs, such as the E-Rate and Rural Health Care programs, explicitly target underserved areas and may channel funds toward locations with limited broadband access. As a result, observed federal outlays to municipalities could partially reflect government efforts to promote broadband diffusion rather than the effect of broadband on political responsiveness.

These considerations motivate an instrumental-variable strategy.

4.1 Instrumental Variable Approach

To address the endogeneity concerns discussed above, I exploit cost-based variation in broadband deployment arising from municipalities' physical proximity to long-haul fiber-optic broadband infrastructure, the Internet *backbone*. This approach follows an established literature that uses technical and geographic barriers to broadband provision as plausibly exogenous sources of variation in broadband diffusion, including distance to backbone infrastructure, terrain characteristics, and other features affecting deployment costs (Barnebeck Andersen et al., 2011; Falck et al., 2014; Miner, 2015; Campante et al., 2018; Gavazza et al., 2019; Hjort and Poulsen, 2019; Amaral-Garcia et al., 2022).

The core intuition is that the cost of providing high-speed Internet increases with distance from backbone infrastructure. In addition to the "last-mile" connection, which links end users to local access points via telephone lines, coaxial cables, or wireless antennas, broadband Internet service providers must connect local networks to the long-haul fiber-optic backbone, a step that involves substantial fixed costs related to excavation, permitting, and installation, typically estimated at \$10,000-\$15,000 per mile (NTIA and RUS, 2000).[7] As a result, municipalities located closer to backbone infrastructure face lower deployment costs, and are more likely to experience earlier and more intensive entry of broadband providers.

Because the exact location of the U.S. broadband Internet backbone is not publicly available due to its high competitive value and security concerns, I follow evidence from network engineering and telecommunications research showing that backbone fiber is frequently deployed alongside

[7]This backbone connection is required for most wireline and wireless broadband technologies. The dominant wireline broadband technologies in the U.S. during the period studied are DSL and cable modem. Wireless broadband refers primarily to mobile technologies.

the National Interstate Highway System (Durairajan et al., 2015). Highways provide shared rights-of-way and lower construction costs, making them a natural conduit for fiber deployment. Importantly, the Interstate Highway System was planned in the early 1950s and largely completed by the late 1980s—well before the emergence of commercial broadband—making its geographic layout predetermined with respect to subsequent broadband demand, contemporaneous local economic shocks, and political outcomes.

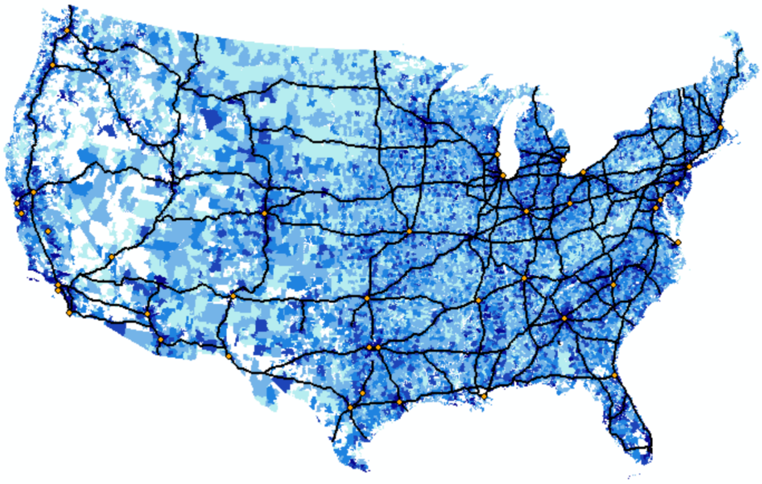
I therefore use proximity to Interstate highways as an instrument for broadband diffusion. Specifically, I measure each municipality's distance to the nearest Interstate highway using road network data from the National Transportation Atlas Database (1995). To mitigate concerns about endogenous routing through urban areas, I exclude auxiliary three-digit Interstate segments added during the 1956 planning revisions.

Given the panel structure of the data, identification does not rely on cross-sectional differences in proximity alone. Since proximity to highways is time invariant, the empirical strategy exploits the fact that the effect of proximity on broadband availability may evolve over time as broadband technologies diffuse and deployment expands outward from backbone infrastructure. Accordingly, I allow the relationship between proximity and broadband diffusion to vary flexibly over time.

Figure 2 provides a visual illustration of this strategy. The map plots the change in the number of ISPs between 1999 and 2010 across ZCTAs, together with the Interstate Highway network.[8] Areas located closer to Interstate highways tend to experience larger increases in ISP presence during the study period, consistent with the cost-based deployment mechanism described above.

[8] I plot ZCTAs instead of municipalities for illustration purposes, since municipalities do not cover the entire U.S. territory.

Figure 2: Entry of ISPs and the Interstate Highway System, 1999-2010



Notes: This map displays the average number of ISPs in ZCTAs over the period 1999–2010. Darker colors represent a higher number of providers. Black lines depict the National Interstate Highway System as of year 1995, and orange dots represent the forty largest U.S. cities in 2000.

In the next subsection, I formalize this intuition and develop the corresponding empirical first-stage specification.

4.1.1 First Stage Model

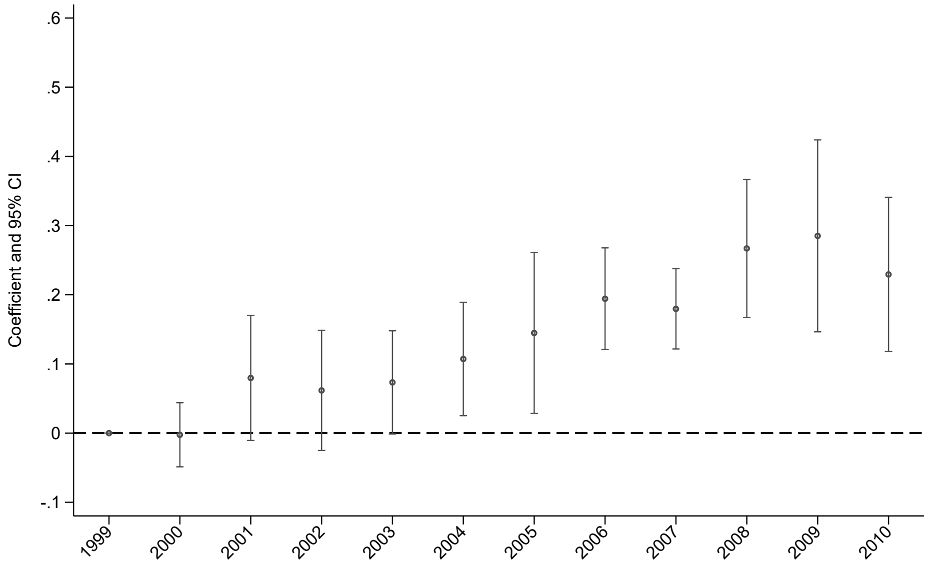
To provide transparency on the source of identifying variation, I begin by estimating a semi-parametric first-stage specification:

$$\text{Number of ISPs}_{mt} = \sum_t \delta_t \text{Proximity}_m \times \mathbb{1}\{year = t\} + X_m^{'} \gamma_t + f_m + f_{st} + u_{mt} \qquad (2)$$

where notation follows Equation (1). Proximity_m measures the municipality's proximity to the Interstate Highway network, defined as the average inverse distance from ZCTA centroids to the nearest Interstate segment, consistent with the aggregation of broadband data. The variable $\mathbb{1}\{year = t\}$ is a year indicator. The coefficients of interest, δ_t, capture how the relationship between proximity to Interstate highways and the number of ISPs evolves over time relative to the baseline year, 1999.

Figure 3 plots the estimated coefficients for δ_t from Equation (2). The estimates show that the association between proximity to Interstate highways and the number of ISPs strengthens steadily over time and follows an approximately linear pattern. This evolution is consistent with broadband markets expanding over the sample period and with provider entry occurring more intensively in locations with lower deployment costs, while more remote municipalities experience slower growth in ISP presence. Appendix C provides additional evidence on the mechanism underlying the first-stage relationship. In particular, it illustrates how proximity to Interstate highways shapes the evolution of ISP presence across municipalities and how these dynamics vary with expansion in the state-level ISP market.

Figure 3: First Stage: Semi-Parametric Specification



Notes: This figure displays estimates of the semi-parametric first-stage specification in Equation (2). The dependent variable is the number of ISPs serving municipality m in year t. The main regressors are the interaction between municipality proximity to Interstate highways and year fixed effects. The figure plots the estimated coefficients δ_t, with 1999 as omitted base year, together with 95% confidence intervals. The regression is weighted by municipality population as of year 2000. The specification includes municipality fixed effects and state-by-year fixed effects, and the full set of baseline controls interacted with a year fixed effect. Robust standard errors clustered at the state level are reported in parentheses.

Motivated by the approximately linear pattern documented above, and for tractability and ease of interpretation, I estimate the first stage using a parametric specification that interacts proximity with a linear time trend:

$$\text{Number of ISPs}_{mt} = \delta(\text{Proximity}_m \times \text{t}) + X_m^{'}\gamma_t + f_m + f_{st} + u_{mt} \tag{3}$$

The coefficient δ captures whether municipalities located closer to backbone infrastructure experience faster entry of broadband providers over time. Given the expansion of broadband markets during the sample period and the presence of substantial fixed deployment costs, this specification allows for differential ISP entry dynamics between locations closer and further to broadband infrastructure.

Because municipalities located near Interstate highways may differ systematically from remote areas, identification relies on a conditional exclusion restriction. Specifically, the identification assumption is that, conditional on baseline municipal characteristics interacted with year effects, municipality fixed effects, and state-year fixed effects, proximity-driven variation in broadband deployment costs affects federal spending only through its impact on the entry of broadband providers. The next section evaluates the relevance of the instrument and provides evidence supporting the plausibility of this identification assumption.

5 Main Results

5.1 First Stage Results

Table 2 reports estimates of the first-stage model at the municipality level (Equation (3)). The dependent variable is number of ISPs providing service in a municipality. Column (1) includes only state-year fixed effects, allowing the raw relationship between proximity to major Interstate highways and broadband provider presence to be identified. The estimates show a positive correlation between proximity and the number of ISPs, which strengthens over time as captured by the interaction between proximity and the time trend. From Column (2) onward, municipality fixed effects are included. Because proximity to Interstate highways is time invariant, its level effect is absorbed by the municipality fixed effects. Therefore, this specification captures whether municipalities located closer to the backbone infrastructure experience faster broadband providers entry. Column (3) further includes baseline sociodemographic, economic, and geographic characteristics interacted with year fixed effects. This is my preferred specification. Although the interaction coefficient declines slightly in magnitude, it remains positive and statistically significant at the 1% level, indicating that proximity to Interstate highways continues to predict ISP entry even after flexibly accounting for differential trends across municipalities with different baseline characteristics.

Table 2: First Stage: Main Specification

Dep. Var: Number of ISPs	(1)	(2)	(3)	(4)	(5)
Log(Proximity to a Major Interstate Highway)	0.4642***				
	(0.0549)				
Log(Proximity to a Major Interstate Highway) × t	0.0537***	0.0537***	0.0259***		
	(0.0090)	(0.0090)	(0.0047)		
Log(Proximity to a Major Interstate Highway) × State Number of ISPs				0.0042***	
				(0.0009)	
Log(Proximity to a Major Interstate Highway) × U.S. Number of ISPs					0.0003***
					(0.0001)
Kleibergen-Paap F-test	106.124	35.966	30.923	23.643	32.094
Municipality FE	-	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓
Controls × Year FE	-	-	✓	✓	✓
Avg Dep. Var. at Baseline	3	3	3	3	3
Avg Num. of ISPs at State/U.S. Level				54	1,211
Municipalities	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064

Notes: This table reports estimates of Equation (3). The dependent variable is the number of ISPs serving municipality m in year t. The main regressor is the interaction between municipality proximity to Interstate highways and a linear time trend. The coefficient on this interaction captures whether municipalities located closer to backbone infrastructure experience higher entry of broadband providers over time. All regressions are weighted by municipality population as of year 2000. The specification includes municipality fixed effects and state-by-year fixed effects, and the full set of baseline controls interacted with a year fixed effect. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

To provide intuition for the magnitude, consider two otherwise similar municipalities, one located 1 kilometer from an Interstate highway and another located 10 kilometers away. Using the estimates from column (3), the difference in the predicted number of ISPs between these municipalities grows gradually over time, reaching approximately one additional ISP by the end of the sample period.[9] This pattern reflects the cumulative nature of broadband entry: municipalities closer to backbone infrastructure do not simply start with more providers, but experience faster growth in provider presence as the market expands.

Finally, columns (4) and (5) show that similar results are obtained when the linear time trend in the instrument is replaced by alternative measures of aggregate broadband diffusion. In these specifications, proximity is interacted with the number of providers operating at the state level or nationally.[10] In all cases, the coefficients remain positive and statistically significant, and are

[9] The difference between these two municipalities in terms of providers is given by $\Delta \text{ISP}_{(A,B),t} = 0.0259 \times (ln(1) - ln(\frac{1}{10})) \times t$. This means that in 2000, both municipalities will have approximately the same number of providers ($\Delta \text{ISP}_{(A,B),2} = 0.0259 \times (ln(1) - ln(0.1)) \times 2 = 0.12$). By the end of the sample, municipality A will have 1 more provider than B ($\Delta \text{ISP}_{(A,B),12} = 0.0259 \times (ln(1) - ln(0.1)) \times 12 = 0.7$).

[10] I use data from the *Internet Access Services Reports* published annually by the FCC on the number of ISPs at the state and U.S. level. This is to avoid over-counting broadband providers that serve in more than one

similar in economic terms although they differ in magnitude. These differences primarily reflect the scale of the interaction term: the number of providers at the state and national levels grows roughly linearly over time but differs substantially in level. Yet, these alternative specifications capture essentially the same underlying variation in broadband diffusion.

5.1.1 Validity of the Instrument

Here, I examine potential threats to the validity of the instrument along the dimensions of relevance and exogeneity. I begin by assessing relevance. In the preferred specification (Column (3)), the Kleibergen-Paap F-statistic is 30.9, exceeding the conventional critical values used to diagnose weak instruments, including the threshold value of 10 (Staiger and Stock, 1997) and the Stock-Yogo critical values for 10% and 15% maximal size distortion in t-tests. This indicates that proximity to Interstate highways interacted with a linear time trend provides strong predictive power for changes in the number of ISPs within municipalities. Moreover, I show that the identifying variation is specific to the Interstate Highway System. In Appendix Figure D.1, I show that proximity to other road networks does not predict differential ISP entry over time, consistent with broadband infrastructure being deployed along Interstates rather than reflecting general transportation access.

I next examine the plausibility of the exclusion restriction. The identifying assumption is that, conditional on municipality and state-year fixed effects and baseline municipal characteristics interacted with year effects, proximity to Interstate highways affects federal spending only through its impact on differential ISP entry. I consider four potential threats to this assumption, which I further develop in Appendix D.

First, proximity may capture pre-existing trends in federal spending unrelated to broadband. I address this concern with reduced-form estimates over the pre-broadband period. Appendix Table D.1 reports parametric estimates using the main specification and shows that all coefficients are small and statistically indistinguishable from zero. Appendix Figure D.2 complements this evidence by presenting semi-parametric estimates, which reveal no differential trends in federal spending prior to the expansion of broadband and show effects emerging only during the 2000s.

Second, highway proximity may be correlated with the presence of military installations and defense contractors, which could directly attract discretionary federal outlays independently of broadband.[11] This concern is particularly relevant in the post-9/11 period, when the *war on*

location.

[11] The system's official name is the *Dwight D. Eisenhower National System of Interstate and Defense Highways*, and national defense was one of the motivations behind the 1956 Federal-Aid Highway Act.

terror led to a substantial expansion in defense spending. Because some municipalities near Interstate Highways host military installations, this expansion could generate differential spending growth correlated with highway proximity. Appendix Table D.2 provides little support for this explanation, as the reduced-form coefficients for defense spending are small and statistically indistinguishable from zero along both the intensive and extensive margins. Moreover, the reduced-form effect on discretionary spending attenuates only slightly and remains positive and significant when I absorb any differential time trend associated with the location of military installations in 1995.

Third, proximity to highways may be correlated with broader changes in local economic conditions, such as market access, labor demand, urban development, or the location of freight and distribution activity, that independently affect federal spending (Michaels, 2008; Duranton, Gilles and Turner, Matthew A., 2012). Although the specification already allows for flexible differential trends in baseline economic and demographic characteristics, I also examine this channel directly. Appendix Table D.3 reports reduced-form estimates using local economic and demographic outcomes as dependent variables. Proximity is associated with small changes in some outcomes, such as population and employment, but these effects are economically negligible relative to their baseline levels. I also provide evidence on the correlation between proximity to highways and changes in freight and distribution activity, since highways lower transport costs for goods and could attract warehousing independently of broadband. I find a small but significant effect on the number of wholesale, transportation, and warehousing establishments, but the discretionary-spending reduced form attenuates only modestly and remains positive and significant when I control for the baseline number of these establishments interacted with year effects.

Proximity also does not predict changes in transportation-related federal spending, the most direct channel through which highway access could affect federal allocations independently of broadband. Finally, the absence of effects on mandatory spending in the main analysis is difficult to reconcile with a mechanism operating through general improvements in local economic conditions, which would affect entitlement-based programs as well. Taken together with the absence of differential pre-trends, this evidence provides little support for the instrument operating through independent changes in local economic conditions.

Fourth, the placement of the Interstate Highway System may reflect political considerations. Some routes were modified to pass through urban areas to secure congressional support, as documented in the Yellow Book (U.S. Bureau of Public Roads, 1955). If these locations retained distinct political characteristics, the exclusion restriction could be violated. I address this concern in two ways. First, when constructing the instrument, I exclude auxiliary three-digit Interstate segments associated with these revisions. Second, I identify municipalities located in Yellow Book target areas and allow them to follow flexible time-varying trends by interacting an indicator for these municipalities with year fixed effects. Appendix Table D.4 shows that the main results are unchanged and, if anything, slightly larger in magnitude, indicating that political targeting of highway placement does not drive the findings.

Taken together, these tests support the relevance of the instrument and are consistent with the plausibility of the exclusion restriction.

5.2 2SLS Results

Table 3 reports OLS and instrumental variable (2SLS) estimates of Equation (1). The dependent variable is the logarithm of per-capita federal spending in municipalities, and the endogenous regressor is the number of high-speed Internet providers serving a municipality. Columns (1)–(3) present OLS estimates under progressively demanding specifications. Once municipality fixed effects and baseline controls are included, the estimated relationship between broadband competition and total spending becomes small and statistically insignificant. Column (4) reports the preferred 2SLS specification, which includes the full set of fixed effects and controls and instruments the number of ISPs with proximity to Interstate highways interacted with time, as described above. The estimated 2SLS coefficient on total spending is positive but imprecisely estimated.

Table 3: The Effect of Broadband Expansion on Federal Spending

Dep. Var.: Log(Spending per Capita)	Total				Mandatory	Discretionary	Loans & Insurance
	OLS	OLS	OLS	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Number of ISPs	0.192***	-0.012*	0.003	0.092	0.018	0.143**	0.093
	(0.021)	(0.007)	(0.005)	(0.055)	(0.075)	(0.064)	(0.058)
Kleibergen-Paap F-test				30.923	30.923	30.923	30.923
Municipality FE	-	✓	✓	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓	✓	✓
Controls × Year FE	-	-	✓	✓	✓	✓	✓
Avg per Capita Spending at Baseline	1,908	1,908	1,908	1,908	97	764	1,047
Municipalities	35,672	35,672	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064	428,064	428,064

Notes: This table reports estimates of the effect of Broadband expansion on municipality federal spending. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t. Columns (1)–(3) report OLS estimates for total federal spending under increasingly demanding specifications. Columns (4)–(7) report 2SLS estimates, where the number of ISPs in a municipality is instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Columns (4)–(7) consider total federal spending, mandatory spending, discretionary spending, and loans and insurance, respectively. All specifications are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. The Kleibergen–Paap F-statistic is reported for the 2SLS specifications. Significance levels: *** 1%, ** 5%, * 10%.

The positive coefficient for total spending masks important heterogeneity across components of federal outlays. Total transfers combine mandatory, discretionary, and loans and insurance spending, which are determined through distinct institutional processes. Mandatory spending is governed by statutory formulas and entitlement rules, leaving limited scope for short-run political discretion. In contrast, discretionary spending is allocated annually through the congressional appropriations process and may therefore be more responsive to changes in information, monitoring, and political engagement. Loans and insurance programs represent another form of federal support that typically requires administrative approval and risk assessment, and may therefore also be sensitive to political incentives.

Columns (5)–(7) decompose federal spending accordingly. Broadband expansion has no statistically significant effect on mandatory spending (column (5)), consistent with the institutional rigidity of entitlement programs. This pattern is inconsistent with a purely mechanical channel in which broadband access simply facilitates take-up of federal programs, since such a channel would predict increases in mandatory transfers where eligibility is formula-based. In contrast, column (6) shows that broadband expansion significantly increases discretionary transfers. The estimated coefficient of 0.143, statistically significant at the 5% level, implies that an additional ISP is associated with approximately a 15% increase in discretionary spending per capita. This is roughly \$115 per resident relative to the baseline level of discretionary spending, \$764. This result is robust across alternative specifications, remaining positive and significant as controls are progressively added (see Appendix Table A.3). Column (7) shows a positive but imprecisely estimated effect on loans and insurance programs, suggesting that broadband diffusion may also influence forms of federal financial support that involve administrative approval or risk allocation and could therefore be more exposed to political incentives.

To examine which policy domains drive the increase in discretionary transfers, Appendix Table A.4 decompose the 2SLS estimates by major budget function. Point estimates are positive across most functions but individually imprecise, consistent with a broad-based increase rather than one concentrated in a single program area. The estimates are qualitatively similar when using a linear probability model for whether a municipality receives any funding in each category (Appendix Table A.5), suggesting the effect could operate partly by extending discretionary transfers to previously unfunded municipalities.

Appendix Table A.6 reports an alternative decomposition of federal spending by object type, which classifies expenditures according to their economic nature (i.e. grants, procurement, and salaries). Within discretionary spending, the estimated effect is concentrated in grants, which account for roughly 60% of discretionary transfers. A smaller but statistically significant effect appears for salaries and wages, which capture the compensation of federal personnel in particular locations rather than politically targeted spending. This category accounts for only about 0.4% of discretionary spending received by municipalities, and excluding it leaves the discretionary-spending effect virtually unchanged (Appendix Table A.7), reassuring that this category does not drive the main results.

Taken together, these results indicate that broadband expansion shifts the composition of federal transfers toward categories subject to discretionary allocation, while leaving formula-based mandatory programs unaffected. The absence of effects on mandatory spending is difficult to reconcile with a purely mechanical increase in program take-up, which would also predict effects on formula-based transfers. Instead, it is consistent with a broader interpretation in which broadband improves citizens' ability to monitor politicians and fosters participation by lowering barriers to engagement. Greater monitoring and participation may, in turn, sharpen electoral incentives in domains where policymakers retain meaningful allocation discretion. The next section examines one implication of this interpretation: whether broadband also increases campaign contributions, a concrete form of political participation.

6 Broadband and Campaign Contributions

In this section, I estimate the effect of broadband expansion on campaign contributions, a concrete and measurable form of political participation. To that end, I use data from the Database on Ideology, Money in Politics, and Elections (DIME) v.4 (Bonica, 2024), which provides a comprehensive record of itemized campaign contributions made by individuals and organizations in U.S. elections from 1979 to 2024. For each contribution, DIME reports the contributor, recipient, amount, party affiliation, and geographic identifiers at the census tract level. Because federal spending decisions are determined by the President and members of Congress, I restrict the analysis to individual contributions to candidates in federal election cycles, including presidential, House, and Senate races, and covering both primary and general elections. The sample is therefore limited to election years between 1996 and 2008, excluding midterm elections. I aggregate contributions to the municipality level and construct two outcome measures: (i) total contributions per capita, and (ii) the number of contributions per capita, both expressed in logarithms.

Table 4 reports 2SLS estimates from my preferred specification. Although the sample is smaller than in the main analysis due to the restriction to election years, the instrument remains strong, with Kleibergen-Paap F-statistics exceeding conventional thresholds discussed in Section 4. Panel A reports results for presidential elections, while Panels B and C report results for House and Senate elections, respectively. For ease of interpretation, coefficients on log per-capita contributions are multiplied by 1,000.

Table 4: The Effect of Broadband Expansion on Campaign Contributions

	Log(Amount Contributed)		Log(Number of Contributions)		
Dep. Var.:	Total	Average	Total	First-time	Small
	(1)	(2)	(3)	(4)	(5)
Panel A. President					
Number of ISPs	0.055***	0.000	1.392**	0.497	0.824*
	(0.019)	(0.001)	(0.660)	(0.329)	(0.479)
Panel B. House of Representatives					
Number of ISPs	0.053	0.001	0.590***	0.119**	0.410***
	(0.032)	(0.001)	(0.212)	(0.059)	(0.139)
Panel C. Senate					
Number of ISPs	0.018	0.001	0.112	-0.038	0.082
	(0.014)	(0.001)	(0.081)	(0.028)	(0.051)
Kleibergen-Paap F-test	22.148	22.148	22.148	22.148	22.148
Municipalities	35,672	35,672	35,672	35,672	35,672
Observations	142,688	142,688	142,688	142,688	142,688

Notes: This table reports 2SLS estimates of the effect of broadband expansion on campaign contributions using the preferred specification. The endogenous regressor is the number of ISPs serving a municipality, instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. The sample is restricted to election years. Panel A reports results for presidential elections, Panel B for House elections, and Panel C for Senate elections. Dependent variables are measured in log per capita terms, and coefficients for the number of campaign contributions are scaled by 1,000 for readability. Regressions are weighted by municipality population as of year 2000. Robust standard errors clustered at the state level are reported in parentheses. The Kleibergen–Paap F-statistic is reported for each specification. Significance levels: *** 1%, ** 5%, * 10%.

The results indicate that broadband expansion increases campaign contributions, although the effects differ across offices and margins of participation. Focusing first on total contributions per capita, the estimated coefficients are positive across all elections, but only precisely estimated for presidential races. In terms of magnitude, an additional ISP increases per-capita contributions to presidential candidates by approximately 6%. The corresponding estimates for House elections are similar in magnitude but not precisely estimated, while no statistically significant effects are detected for Senate elections.

To better understand the margins of participation, I distinguish between intensive and extensive margins of participation. Column (2) shows that broadband expansion does not significantly affect the average contribution size, indicating that the increase in total contributions is unlikely to be driven by larger donations from existing contributors. Instead, columns (3) to (5) show that broadband expansion increases the number of contributions per capita, pointing to an expansion in participation. In particular, column (4) focuses on first-time contributors, defined as individuals who had not previously made an itemized contribution in the DIME data. The estimates indicate that about 20% of the increase in contributions is driven by these new participants in House elections. This is a central finding in this paper, and suggests that broadband expansion attracted new donors rather than simply intensifying giving among existing donors.

This extensive-margin result relates to Petrova et al. (2021), who find that politicians adopting Twitter attract new donors to their own campaigns. Because both their variation and their outcome are defined at the candidate level, their result captures individual candidates expanding their donor base, a pattern also consistent with donors reallocating toward technology-adopting politicians. My design identifies first-time contributors at the municipality level, and thus isolates entry into campaign finance by new participants from a redistribution of existing political giving across candidates.

I next examine whether these effects are concentrated among small contributions, defined as donations below $200. This margin has received substantial attention in U.S. elections, as the rise of small-dollar donors, in part through online fundraising platforms such as ActBlue and WinRed, has been interpreted as evidence of broader political participation and lower barriers to entry into campaign finance (Bouton et al., 2022). Consistent with this mechanism, I evaluate whether broadband expansion increases small contributions. The estimates indicate that it does so in both presidential and House elections (column (5)). Together with the increase in first-time donors, these findings suggest that broadband expansion plausibly lowered barriers to entry into campaign finance, enabling new participants to engage at relatively low contribution levels.

The results on small contributions should be interpreted with caution. The DIME data include only itemized contributions reported to the Federal Election Commission, such that only donors contributing more than $200 in aggregate over an election cycle are individually identified. As a result, the small contributions observed in the data come from donors who exceed this threshold in total, and therefore do not capture the full set of small-dollar donors. Consequently, the measure of small contributions in this analysis reflects the lower tail of itemized donations rather than the universe of small-dollar contributions. The estimates thus should be interpreted as a lower bound of the effect of broadband expansion on small-scale political participation.

The absence of precisely estimated effects in Senate elections may reflect institutional features of these races. Senate elections are less frequent and geographically broader, with only a subset of states holding elections in each cycle, which reduces variation at the municipality level and limits statistical power. Reduced-form estimates are reported in Appendix Figure D.4, which show no evidence of differential pre-trends and document an increase in contributions from the 2000 election cycle in municipalities closer to the backbone.

Taken together, these results indicate that broadband expansion increased political engagement through campaign contributions, both by raising contribution intensity and expanding participation. This rise in participation provides a plausible channel through which broadband expansion affects discretionary federal spending. Consistent with recent evidence that digital technologies improve political accountability and intensify electoral competition (Guriev et al., 2021, 2025; Petrova et al., 2021; Enríquez et al., 2024), these findings suggest that improved information environments can broaden political participation and strengthen electoral incentives. In the U.S. context, where campaign finance plays a central role in electoral competition, broadband expansion is likely to have increased the political salience of municipalities by enlarging their donor base and lowering the costs of political participation. While these findings do not by themselves isolate how broadband affects federal spending, they identify a plausible channel operating through political participation.

7 Conclusion

This paper studies how the diffusion of broadband Internet affects government responsiveness by improving political engagement and, in turn, strengthening electoral incentives. Using a cost-based identification strategy based on proximity to the Interstate Highway System, I show that broadband expansion led to an increase in discretionary transfers while leaving mandatory, formula-based programs unaffected. This pattern points to a reallocation of public resources toward areas where political responsiveness is most relevant.

Consistent with this interpretation, I show that broadband expansion increased political engagement through increased campaign contributions in presidential and congressional elections. These effects are driven in part by the entry of new donors and the expansion of small contributions, suggesting that broadband Internet access lowered barriers to participation and broadened engagement in the political process. While this evidence does not by itself isolate the channel linking broadband to federal spending, it identifies political participation as a plausible one, and one difficult to reconcile with a purely mechanical account of program take-up.

Taken together, these findings suggest that, in its early stages, broadband diffusion strengthened electoral incentives by expanding participation in the political process. More broadly, they help reconcile mixed evidence in the literature: while broadband access may crowd out political engagement in some settings, in environments characterized by higher information frictions and meaningful policy discretion it can instead enhance political responsiveness. Whether these effects persist in more mature, high-information environments, where online content and modes of engagement have evolved substantially, remains an open question for future research.

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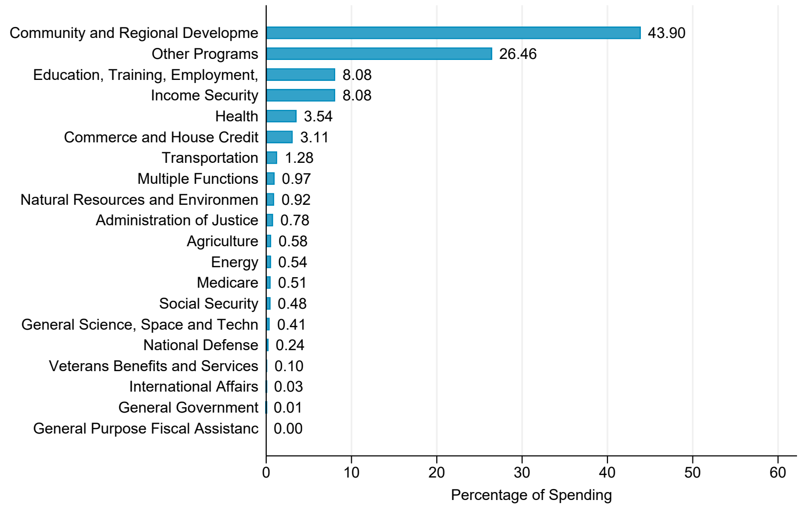
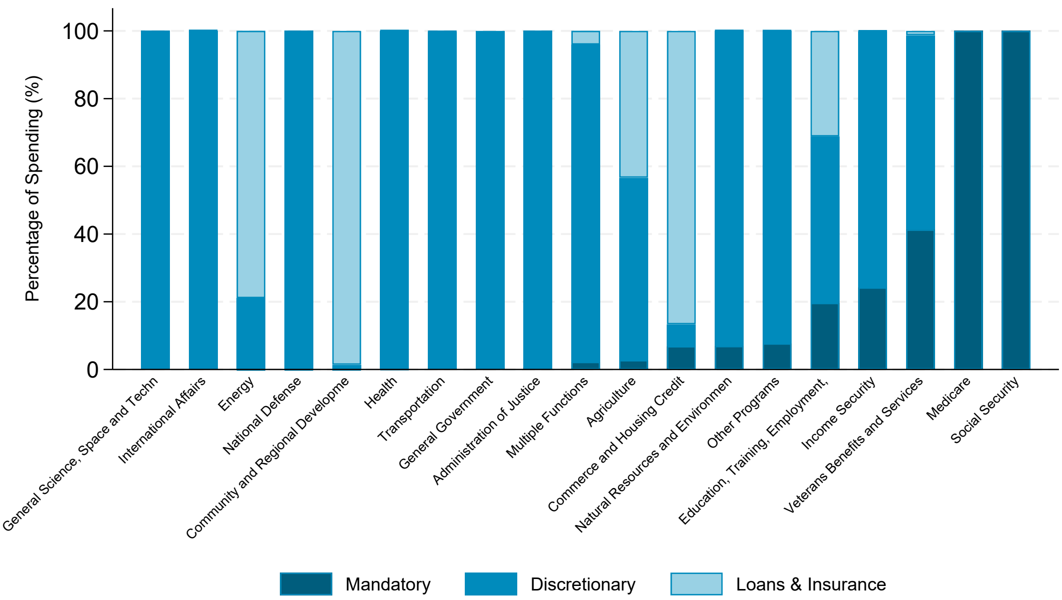
A Additional Tables and Figures

Figure A.1: Composition of Federal Spending by Major Budget Function



Notes: The figure shows the composition of federal spending by major object functions. For each municipality-year in 1999–2010, I compute the share of spending in each function category relative to total spending. Bars report the population-weighted mean of these municipality-year shares, using municipality population in 2000 as weights.

Figure A.2: Composition of Federal Spending by Budget Function and Spending Type



Notes: This figure shows the composition of federal spending by major budget function and spending type. For each municipality-year in 1999–2010, I compute the share of spending accounted for by mandatory, discretionary, and loans and insurance within each major budget function. Bars report the population-weighted mean of these shares, using municipality population in 2000 as weights.

Table A.1: Baseline Municipality Characteristics by Broadband Availability and Number of ISPs

	Broadband Availability				*Number of ISPs (cond. on Covered)*			
	Uncovered	Covered	Diff.	P-value	$\leq$ Median	$>$ Median	Diff.	P-value
Log of Total Population	8.020	9.894	1.874	0.000	9.641	10.269	0.629	0.000
Log of Population Density	2.944	5.746	2.802	0.000	4.913	6.980	2.067	0.000
Share of Rural Population	0.799	0.196	-0.603	0.000	0.301	0.041	-0.260	0.000
Log(Elevation)	5.361	4.855	-0.507	0.000	5.102	4.489	-0.613	0.000
Median Income	36,010	46,772	10,762	0.000	42,668	52,850	10,182	0.000
Share of Graduates	0.080	0.138	0.058	0.000	0.118	0.168	0.049	0.000
Share of Population over 65	0.146	0.126	-0.020	0.000	0.132	0.117	-0.014	0.000
Share of Black	0.060	0.111	0.052	0.000	0.096	0.133	0.037	0.005
Share of Hispanic	0.049	0.108	0.059	0.000	0.076	0.156	0.080	0.000
Share of Veterans	0.140	0.126	-0.014	0.000	0.136	0.112	-0.025	0.000
Unemployment Rate	0.056	0.057	0.000	0.780	0.057	0.057	0.001	0.773
Share Employment in Farming	0.025	0.007	-0.019	0.000	0.010	0.002	-0.007	0.000
Share of Government Employees	0.151	0.143	-0.008	0.000	0.148	0.137	-0.011	0.009
Log(Proximity to State Capital)	-4.914	-4.690	0.224	0.000	-4.731	-4.629	0.103	0.236
Log(Proximity to Coast)	-5.233	-4.232	1.000	0.000	-4.640	-3.629	1.011	0.000
Log(Proximity to Main City)	-5.262	-4.275	0.987	0.000	-4.838	-3.440	1.398	0.000

Notes: This table reports population-weighted means of baseline municipality characteristics in 1999 by Broadband availability and ISP presence. Columns (1) to (4) compare municipalities with no ISP presence (*Uncovered*) to municipalities with at least one ISP offering service (*Covered*). Columns (5) to (8) restrict the sample to covered municipalities and compare municipalities with a number of ISPs below the median to those above the median. The table reports differences in means across groups together with the corresponding p-values. All observations are weighted by municipality population in 2000.

Table A.2: Summary Statistics of Selected Municipality Characteristics

	Mean	SD	Min	Max
Total Population	9,044.699	61,077.456	9	8,009,843
Population Density	2,001.054	4,333.418	0	201,026
Share of Population over 65	0.127	0.043	0	1
Share of Black	0.108	0.148	0	1
Share of Hispanic	0.104	0.148	0	1
Share of Veterans	0.127	0.033	0	0
Share of Graduates	0.135	0.065	0	0
Median Income	46,109.643	15,018.045	6,250	200,001
Unemployment Rate	0.057	0.028	0	1
Share Employment in Farming	0.008	0.018	0	1
Share of Government Employees	0.144	0.052	0	1
Share of Rural Population	0.233	0.307	0	1
Elevation	256.525	325.693	0	3,603
Proximity to State Capital	0.020	0.044	0	1
Proximity to Coast	0.090	2.241	0	512
Proximity to Main City	0.031	0.044	0	0

Notes: This table reports population-weighted means of the baseline municipality characteristics used in the analysis. Population density, elevation, median income, and proximity measures are reported in levels rather than logarithms to ease interpretation. All calculations use municipality population in 2000 as weights.

Table A.3: The Effect of Broadband Expansion on Discretionary Spending: Alternative Specifications

Dep. Var.: Log(Discretionary Spending per Capita)	OLS (1)	OLS (2)	2SLS (3)	2SLS (4)	2SLS (5)
Number of ISPs	0.009	0.013***	0.143**	0.127*	0.143*
	(0.006)	(0.005)	(0.064)	(0.068)	(0.076)
Kleibergen-Paap F-test			30.923	31.976	20.872
Municipality FE	✓	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓
Controls × Year FE	-	✓	✓	✓	✓
Prox. to Military Bases × Year FE	-	-	-	✓	-
Freight & Distrib. Establishments × Year FE	-	-	-	-	✓
Avg per Capita Spending at Baseline	764	764	764	764	764
Municipalities	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064

Notes: This table reports OLS and 2SLS estimates of the effect of broadband expansion on discretionary federal spending under increasingly demanding specifications. The dependent variable is the logarithm of one plus discretionary federal spending per capita in municipality m and year t, and the endogenous regressor is the number of ISPs serving a municipality. Columns (1) and (2) report OLS estimates, without and with baseline controls interacted with year effects. Columns (3)–(5) report 2SLS estimates, where the number of ISPs is instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Column (3) is the preferred specification; columns (4) and (5) add baseline proximity to military installations and the baseline number of freight and distribution establishments, respectively, each interacted with year effects. All regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Table A.4: The Effect of Broadband Expansion on Discretionary Spending, by Function

Dep. Var: Log(Spending per Capita)	Community and Reg. Development (1)	Income Security (2)	Educ., Empl., and Soc. Services (3)	Health (4)	Commerce and House Credit (5)	Other Spending (6)
Number of ISPs	-0.062	0.167	0.048	-0.023	0.054	0.013
	(0.044)	(0.113)	(0.040)	(0.055)	(0.054)	(0.058)
Kleibergen-Paap F-test	30.923	30.923	30.923	30.923	30.923	30.923
Share over Disc. Spending at Baseline	0.040	0.287	0.082	0.060	0.002	0.530
Municipalities	35,672	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064	428,064

Notes: This table reports 2SLS estimates of the effect of broadband expansion on discretionary federal spending by major budget function using the preferred specification. The dependent variable is the logarithm of one plus discretionary federal spending per capita in municipality m and year t. The endogenous regressor is the number of ISPs serving a municipality, instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Table A.5: The Effect of Broadband Expansion on Federal Spending, by Function (Extensive Margin)

Dep. Var: 1{Spending > 0 }	Community and Reg. Development (1)	Income Security (2)	Educ., Empl., and Soc. Services (3)	Health (4)	Commerce and House Credit (5)	Other Spending (6)
Number of ISPs	-0.008	0.028	0.007	-0.039*	-0.023	-0.018**
	(0.016)	(0.029)	(0.019)	(0.023)	(0.023)	(0.009)
Kleibergen-Paap F-test	30.923	30.923	30.923	30.923	30.923	30.923
Avg Dep. Var. at Baseline	0.505	0.726	0.564	0.449	0.250	0.762
Municipalities	35,672	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064	428,064

Notes: This table reports 2SLS estimates of the effect of broadband expansion on the probability that a municipality receives discretionary federal funding by major budget function, using the preferred specification. The dependent variable is an indicator equal to one if municipality m receives positive discretionary spending in a given function in year t. The endogenous regressor is the number of ISPs serving a municipality, instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Table A.6: The Effect of Broadband Expansion on Discretionary Spending, by Object Type

Dep. Var.: Log(Spending per Capita)	Grants (1)	Procurement Contracts (2)	Salaries (3)
Number of ISPs	0.121*	0.075	0.344***
	(0.072)	(0.066)	(0.083)
Kleibergen-Paap F-test	30.923	30.923	30.923
Share over Disc. at Baseline	0.604	0.391	0.004
Municipalities	35,672	35,672	35,672
Observations	428,064	428,064	428,064

Notes: This table reports 2SLS estimates of the effect of broadband expansion on discretionary spending by object type using the preferred specification. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t. The endogenous regressor is the number of ISPs serving a municipality, instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Object types classify expenditures according to their economic nature (grants, procurement, and salaries and wages). Regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Table A.7: The Effect of Broadband Expansion on Federal Spending, excl. Salaries of Federal Employees

Dep. Var: Log(Spending per Capita)	Total				Mandatory	Discretionary	Loans & Insurance
	OLS	OLS	OLS	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Number of ISPs	0.191***	-0.014**	0.003	0.096*	0.018	0.141**	0.093
	(0.021)	(0.007)	(0.005)	(0.056)	(0.075)	(0.066)	(0.058)
Kleibergen-Paap F-test				30.923	30.923	30.923	30.923
Municipality FE	-	✓	✓	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓	✓	✓	✓
Controls × Year FE	-	-	✓	✓	✓	✓	✓
Avg per Capita Spending at Baseline	1,908	1,908	1,908	1,908	97	764	1,047
Municipalities	35,672	35,672	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064	428,064	428,064

Notes: This table reports 2SLS estimates of the effect of broadband expansion on federal spending excluding salaries and wages of federal employees, using the preferred specification. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t, by type of spending. The endogenous regressor is the number of ISPs serving a municipality, instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. Regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

B Number of ISPs as a Proxy for Broadband Internet Subscriptions

Detailed data on broadband subscriptions at fine geographic levels in the U.S. are not available prior to 2008. As a result, I use the number of broadband Internet service providers (ISPs) operating in a municipality as a proxy for broadband take-up. This appendix provides institutional context for this choice and presents suggestive evidence supporting its empirical relevance.

Decisions by ISPs regarding where to offer broadband services reflect both the costs of deployment and expected local demand. Historically, entry into U.S. telecommunications markets was shaped by regulatory constraints and legacy infrastructure, with early broadband provision dominated by licensed local monopolies that owned telephone local exchanges and copper lines used to provide DSL, one of the main broadband technologies. The Telecommunications Act of 1996 expanded entry opportunities by allowing new firms to build facilities or lease existing infrastructure, contributing to broader broadband deployment during the late 1990s and 2000s.

Descriptive evidence from regulatory agencies indicates that areas with more providers tended to experience greater broadband availability, lower prices, and higher service quality, factors commonly associated with broadband adoption (NTIA and RUS, 2000; NTIA, 2007, 2013). While these patterns are not causal, they suggest that variation in the number of ISPs captures meaningful differences in local broadband market conditions relevant for broadband take-up.

I complement this discussion with empirical evidence using FCC Form 477 data for the period 2008–2013, when information on both residential fixed broadband subscriptions and the number of ISPs is available at fine geographic levels. Residential fixed broadband subscriptions are reported in categorical bins indicating the number of connections per 1,000 households. I convert this variable into a continuous measure by assigning each category its midpoint and expressing subscriptions as a rate. The data are aggregated to the municipality level.

Table B.1 reports OLS estimates from regressions of the constructed subscription rate on the average number of ISPs serving a municipality, progressively adding fixed effects and a rich set of sociodemographic, economic, and geographic controls.[12] Across specifications, there is a robust positive association between ISP presence and residential broadband adoption. In the most demanding specification, an additional ISP is associated with a 0.2 percentage point increase in the household subscription rate. In terms of magnitude, a one–standard deviation increase in the

[12] The instrumental variable used in the main analysis lacks sufficient power over this period. I therefore focus on OLS relationships in this validation exercise.

number of providers (approximately four ISPs) corresponds to a 0.8 percentage point increase, or about 1.4% relative to the mean household subscription rate at baseline. This relationship is stable across specifications, alleviating concerns that ISP counts simply proxy for observable local characteristics.

Table B.1: Correlation Number of ISPs and Subscriptions

Dep. Var: Residential subscription rate	(1)	(2)	(3)	(4)
Number of ISPs	-0.001	0.002***	0.002***	0.016
	(0.001)	(0.001)	(0.001)	(0.029)
× Share rural				0.004**
				(0.002)
× Log median HH income				-0.001
				(0.003)
× Share Graduates				-0.018
				(0.013)
Municipality FE	✓	✓	✓	✓
State × Year FE	-	✓	✓	✓
Controls × Year FE	-	-	✓	✓
Avg Subscription Rate in 2008	0.60	0.60	0.60	0.60
Municipalities	35,672	35,672	35,672	35,672
Observations	214,032	214,032	214,032	214,032

Notes: This table reports OLS regressions of residential fixed broadband subscription rates on the number of ISPs serving a municipality. Subscription data come from FCC Form 477 for the period 2008–2013 and are reported in categorical bins indicating ranges of connections per 1,000 households. I convert these categories into a continuous measure by assigning each bin its midpoint and expressing subscriptions as a rate. All regressions are weighted by municipality population. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

While modest in magnitude, this association is consistent with the stage of broadband diffusion during the period under analysis. By the late 2000s, broadband availability was already widespread: more than 90% of the U.S. population lived in municipalities served by at least one provider, and roughly 57% of households had broadband access at home. In this context, additional provider entry is unlikely to generate large increases in aggregate take-up, but rather to reflect incremental adoption among late adopters or switching across providers. Column (4) supports this interpretation: the association between ISP presence and subscriptions is stronger in more rural municipalities, where broadband adoption initially lagged behind. The estimates also suggest weaker associations in municipalities with higher household income and a larger share of college-educated residents, although these differences are not statistically significant.

These patterns are consistent with findings in the telecommunications and industrial organization literature. For example, Wallsten and Mallahan (2010) documents a negative association between broadband prices and provider competition in U.S. markets, while Nardotto et al. (2015) shows that forcing incumbent telephone operator to make the local telephone network (used in DSL technologies) available to new entrants encouraged entry and substantially improved broadband penetration in early years. Related evidence indicates that more competitive broadband markets are associated with higher average service speeds (Molnar and Savage, 2017). Together with the empirical evidence documented here, this body of literature supports the use of the number of ISPs as an informative proxy for broadband adoption when direct subscription data are unavailable.

C Additional Evidence on the First Stage Mechanism

This section provides additional evidence on the mechanism underlying the first-stage relationship between proximity to Interstate highways and the number of broadband providers documented in Section 5.1. In particular, the results show that municipalities located closer to backbone infrastructure systematically accumulate more providers over time. To illustrate this mechanism more transparently, I present a complementary exercise, where I discretize the number of ISPs to examine how proximity affects the distribution of ISP presence across municipalities over time.

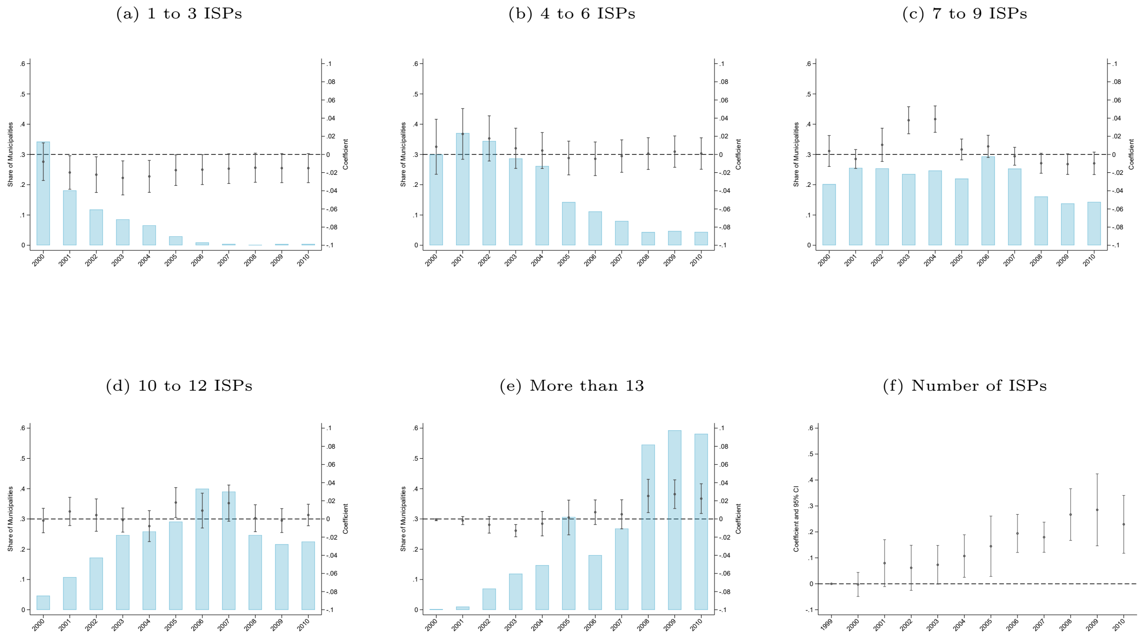
More specifically, I discretize the number of ISPs into ranges and estimate how proximity predicts the probability that a municipality falls within each range in a given year. The estimated specification is:

$$\mathbb{1}\{a \leq \text{Number of ISPs} \leq b\}_{mt} = \delta_t \text{Proximity}_m + X_m^{'} \gamma_t + f_m + f_{st} + u_{mt} \tag{4}$$

The dependent variable is an indicator equal to one if municipality m in year t has between a and b broadband providers. By interacting proximity with year fixed effects, the specification allows the relationship between proximity and ISP presence to vary flexibly over time. The coefficients δ_t therefore capture how proximity to backbone infrastructure shifts municipalities across different levels of ISP presence in each year of the sample.

Figure C.1 displays the results. Panels (a) to (e) report the annual regression coefficients for each of the dummy variables. The blue bars represent the average of the dependent variable for every year. Panel (f) uses as dependent variable Number of ISPs. The first thing to notice is that the distribution of providers in a municipality, captured by the blue bars, shifts to the right over time. That is, over time the share of municipalities with low levels of providers decrease in favor of more providers. The second thing to notice is that the growth in the share of municipalities in each panels is firstly driven by the ones closer to the broadband infrastructure, and eventually further away localities catch up. However, when they catch up, closer municipalities already have a higher number of providers. That is, ISPs first provide broadband Internet where the cost is lower.

Figure C.1: First Stage: Discretization of the Number of ISPs



Notes: This figure reports estimates from semi-parametric regressions of Equation (4). Panels (a)–(e) use as the dependent variable a dummy equal to one if the number of ISPs in a municipality falls within a given range (e.g., 1–3, 4–6, etc.). The blue bars show the mean of each indicator, corresponding to the share of municipalities in each ISP-count category. Panel (f) uses the number of ISPs in a municipality as the dependent variable, as in Figure 3. All regressions are weighted by municipality population. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Nonetheless, this is not enough to explain the increase in importance of proximity for attracting providers over time. For that, it is necessary that ISP entry in closer municipalities is faster than the rate at which further away municipalities are catching up in terms of providers. Start comparing panels (a) and (b). In early years, the coefficients of proximity cancel out, that is, entry rates are similar for both close and faraway municipalities. However, as one can observe in panel (c), as the market continues growing, further away municipalities are not able to absorb new providers at the same rate. In later years the market start decelerating, but still nearby municipalities are the ones that are able to attract a very high number of providers (more than 13). This is reflected in positive coefficients from year 2008 but a flat trend.

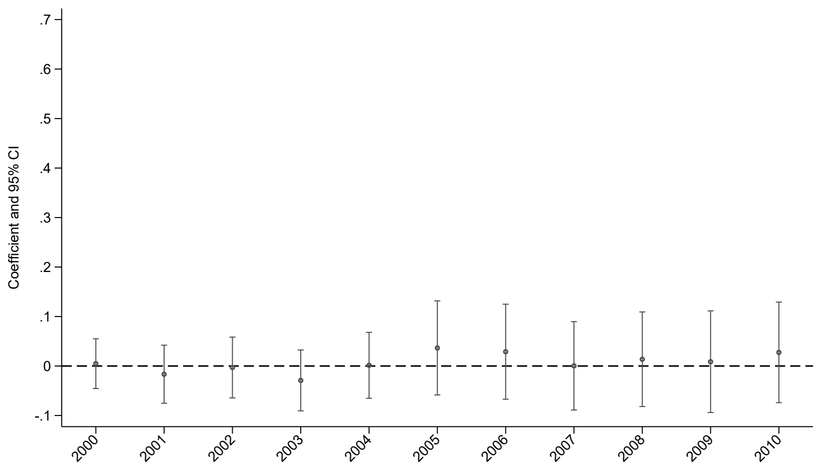
Overall, these figures show that in early years, faraway places catch up at similar rates than the rate at which closer municipalities grow in terms of providers. However, over time ISPs continue entering the markets where cost of service is lower, while entry in markets farther away from broadband infrastructure stagnates.

D Additional Evidence on Instrument Validity

D.1 Relevance

Figure D.1 reports semi-parametric first-stage estimates replacing proximity to Interstate highways with proximity to all other roads in the 1995 National Transportation Atlas, excluding the Interstate system. The coefficients are flat and statistically indistinguishable from zero throughout the sample period, confirming that the identifying variation operates specifically through broadband infrastructure co-deployed along Interstate rights-of-way rather than through general transportation access.

Figure D.1: First Stage: Proximity to all roads, excl. Interstate Highways



Notes: This figure displays the semi-parametric estimation of the first stage at the municipality level. The dependent variable is Number of ISPs in a municipality. The right-handside variable is proximity to any road, excluding Interstate highways. All regressions are weighted by municipality population. Robust standard errors are clustered by state. Each annual coefficient is reported, as well as 95% confidence intervals.

D.2 Exogeneity

Pre-existing Trends in Federal Spending

Table D.1 reports reduced-form estimates over the pre-broadband period 1994–1998, regressing federal spending on proximity interacted with a linear time trend, using my main specification. The point estimate is small and negative, though not statistically distinguishable from zero, indicating that if anything the differential in federal spending by proximity was declining over the pre-broadband period.

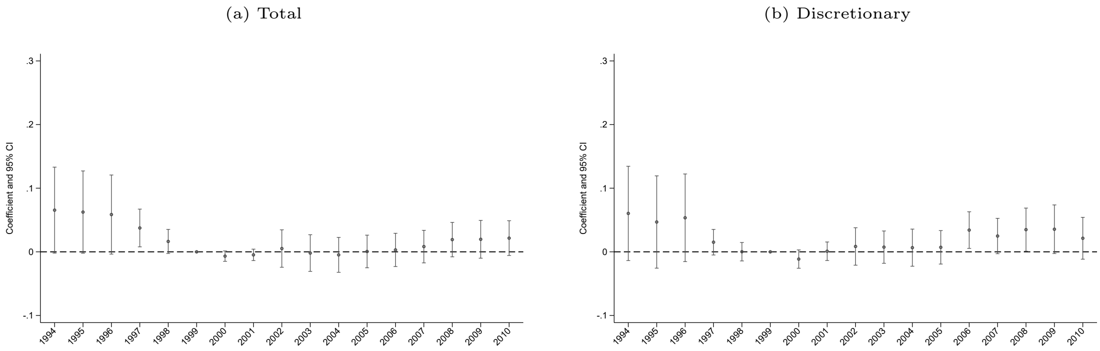
Table D.1: Reduced-Form Effects of Proximity on Federal Spending

Dep. Var.: Log(Spending per Capita)	Total	Mandatory	Discretionary	Loans & Insurance
	(1)	(2)	(3)	(4)
Log(Proximity to a Major Interstate Highway) $\times$ t	-0.012	-0.008	-0.015	-0.004
	(0.008)	(0.005)	(0.010)	(0.005)
Avg per Capita Spending at Baseline	1,573	30	872	671
Municipalities	35,672	35,672	35,672	35,672
Observations	178,360	178,360	178,360	178,360

Notes: This table reports reduced-form estimates of the effect of Proximity times a linear trend on municipality federal spending during the pre-broadband period 1994-1998. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t. All specifications are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Figure D.2 complements this evidence with the semi-parametric specification over the full sample, plotting the reduced-form coefficients from 1994 onward. The coefficients are flat for discretionary spending prior to the broadband era and increase only as broadband Internet access expands during the 2000s, when broadband was first made commercially available. Over the pre-broadband period the coefficients are positive but estimated with wide confidence intervals, which could raise a concern about differential pre-trends. Two features of the estimates alleviate this concern. First, the coefficients in the years immediately preceding 1999, the base year, are precisely estimated and close to zero. Formally, I cannot reject the null that the 1997 and 1998 coefficients are jointly zero ($F(2, 47) = 1.51$, $p - value = 0.23$). Since these are the observations closest to the onset of broadband, they are the most informative about whether spending was already diverging by proximity before treatment began, and they show no such divergence. Second, the pre-period coefficients decline toward zero as they approach the 1999 base year, consistent with the negative linear-trend estimate in Table D.1. Any differential trend they might reflect therefore runs opposite to the increase estimated during the broadband era. A pre-existing trend of this kind would bias the main estimate toward zero rather than away from it, and so cannot account for the positive effect I find.

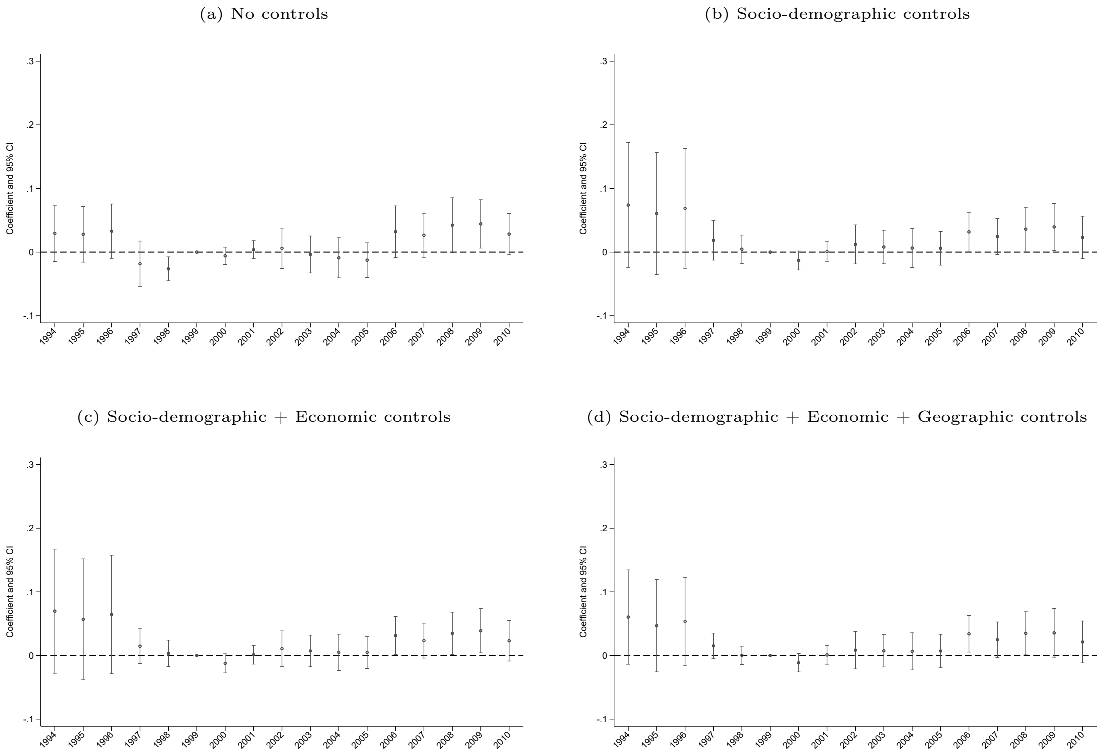
Figure D.2: Reduced-Form Effects of Proximity on Federal Spending (Semi-parametric)



Notes: This figure reports reduced-form estimates of the effect of proximity to Interstate highways on federal spending using the preferred specification. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t. Panel (a) considers total federal spending, while panel (b) considers discretionary spending. The coefficients shown correspond to interactions between municipality proximity to an Interstate highway and year indicators. The figure plots the estimated coefficients together with 95% confidence intervals. All regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are used to construct confidence intervals.

Finally, I assess whether the reduced-form pattern on discretionary spending depends on the choice of controls. Starting from a specification with only municipality and state-year fixed effects, I progressively add sociodemographic, economic, and geographic controls. Appendix Figure D.3 plots the coefficients under each. The post-1999 pattern is unchanged throughout: the effect emerges only in the 2000s and remains in the same range regardless of the controls included. The pre-1999 coefficients are equally stable once the first block of controls is included, and adding the economic and geographic controls leaves them essentially unchanged, although the confidence intervals in the earliest pre-period years widen. The reduced-form pattern is therefore robust to the choice of controls.

Figure D.3: Reduced-Form Effects on Discretionary Spending - Sensitivity to Controls Choice



Notes: This figure reports semi-parametric reduced-form estimates of the effect of proximity to Interstate highways on log discretionary federal spending per capita, under specifications that progressively add controls. The coefficients correspond to interactions between municipality proximity to an Interstate highway and year indicators, relative to the 1999 base year. Panel (a) includes only municipality and state-year fixed effects. Panel (b) adds socio-demographic controls: log population, log population density, share of rural population, median income, share of graduates, share of population over 65, share of Black population, share of Hispanic population, and share of veterans. Panel (c) adds economic controls: the unemployment rate, share of employment in farming, and share of government employees. Panel (d) adds geographic controls: log mean elevation, proximity to the coast, and proximity to the state capital and to a major city. All regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are used to construct 95% confidence intervals.

Defense Spending and the War on Terror as a Confounding Channel

A concern for the exclusion restriction is that proximity to Interstate highways is correlated with the presence of military installations and defense contractors. The system was conceived partly for national defense, and highway-proximate municipalities may host military facilities and the contractors that serve them. If so, the expansion of defense spending during the *war on terror* after 9/11 could generate differential growth in discretionary federal spending in these municipalities for reasons unrelated to broadband. Under this mechanism, proximity interacted with a time trend would predict differential growth in defense outlays. I test this directly by re-estimating the reduced form with defense spending as the outcome, and by controlling for the geography of military installations.

Table D.2 reports the results. Column 1 reproduces the reduced-form effect on log discretionary spending per capita, the main outcome. Columns 2 and 3 replace the outcome with log defense spending per capita and with an indicator for any defense procurement. In both cases the coefficient is small and statistically indistinguishable from zero, so proximity predicts differential growth in defense spending on neither the intensive nor the extensive margin. Because defense spending is itself a component of discretionary spending, its reduced-form coefficient also bounds the share of the total effect that defense can mechanically account for. Defense procurement represents roughly 34% of discretionary spending at baseline. At the point estimate, it accounts for approximately $0.34 \times 0.001/0.004 \approx 9\%$ of the estimated effect on discretionary spending. Column 4 reproduces the main specification and adds the interaction of proximity to military installations in 1995 with year fixed effects, which absorbs any differential time path associated with the geography of defense activity. The estimate attenuates only slightly and remains positive and significant. Together, these results provide little support for the concern that the estimated effect on discretionary spending operates only through defense allocations to municipalities.

Table D.2: Reduced-Form Effects of Proximity on Defense Spending

Dep. Var.:	Log(Discretionary Spend. per Capita) (1)	Log(Defense Spending per Capita) (2)	Any Defense Contract (3)	Log(Discretionary Spend. per Capita) (4)
Log(Proximity to a Major Interstate Highway) × t	0.004**	0.001	0.000	0.003*
	(0.002)	(0.002)	(0.000)	(0.002)
Avg DV at Baseline (without logs)	764.00	260.00	0.65	764.00
Municipalities	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064

Notes: This table reports reduced-form estimates of the effect of proximity times a linear trend on federal spending. In column (1), the dependent variable is the logarithm of one plus discretionary federal spending per capita, the main outcome. In column (2), it is the logarithm of one plus defense spending per capita. In column (3), it is an indicator for any defense procurement in a municipality. Column (4) reproduces the specification in column (1), adding the interaction of proximity to military installations in 1995 with year fixed effects. All specifications are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Local Economic Conditions and Transportation Spending as a Confounding Channel

A further concern is that proximity to Interstate highways is correlated with broader local economic conditions, or with the location of freight and distribution activity, that could affect federal spending independently of broadband. To examine this channel directly, I estimate reduced-form specifications using a set of local economic and demographic characteristics, freight-sector activity, and transportation-related federal spending as dependent variables over the period 1999–2010. In particular, I use data on population and the share of disadvantaged population from the National Neighborhood Data Archive (NaNDA), and on the number of employees and establishments from the ZIP Code Business Patterns dataset. Table D.3 reports the results.

Table D.3: Reduced-Form Effects of Proximity on Local Economic Conditions

Dep. Var.:	Population	Share Disadvantaged	Number of Employees	Number of Establishments	Transportation Spending	Log(Discretionary Spend. per Capita)
	(1)	(2)	(3)	(4)	(5)	(6)
Log(Proximity to a Major Interstate Highway) $\times$ t	0.001**	-0.000	0.002**	0.002***	0.000	0.003*
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
Avg DV at Baseline (without logs)	417,369.00	0.08	160,883.00	1,062.00	25.00	764.00
Municipalities	35,672	35,672	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064	428,064	428,064

Notes: This table reports reduced-form estimates of the effect of proximity times a linear trend on local economic conditions and federal spending. The dependent variable is the logarithm of one plus population in column (1), the share of disadvantaged population in column (2), the logarithm of one plus the number of employees in column (3), and the logarithm of one plus the number of establishments in the wholesale, transportation, and warehousing sectors (NAICS 42 and 48–49) in column (4). In column (5) it is the logarithm of one plus federal spending on transportation-related programs per capita. Column (6) reports the reduced-form effect on the logarithm of one plus discretionary spending per capita, controlling for the baseline number of wholesale, transportation, and warehousing establishments interacted with year effects. All specifications are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. Significance levels: *** 1%, ** 5%, * 10%.

Proximity is associated with small changes in some local outcomes, including population and the number of employees, but these effects are economically modest and do not follow a systematic pattern across the variables plausibly related to federal spending. For instance, a 1% increase in proximity to a major Interstate highway is associated with an increase of $0.002 \times t\%$ in the number of employees, where t indexes years since the beginning of the sample. Even at the end of the period, the implied effect is approximately 0.02%. I pay particular attention to freight and distribution activity, since highways lower transport costs for goods and could attract warehousing independently of broadband. Column 4 shows a small but significant effect on the number of establishments in the wholesale, transportation, and warehousing sectors. To assess whether this channel drives the main result, column 6 re-estimates the reduced form on discretionary spending, controlling for the baseline number of these establishments interacted with year fixed effects. The estimate attenuates modestly and remains positive and significant, indicating that the effect does not operate through the location of freight activity. Proximity also does not predict changes in transportation-related federal spending (column 5), the most direct channel through which highway access could affect federal allocations independently of broadband.

Finally, the absence of effects on mandatory spending in the main analysis is difficult to reconcile with a channel operating through general improvements in local economic conditions, which would be expected to affect a broader set of federal transfers. Taken together with the pre-trends analysis, these findings provide little evidence that the instrument operates through independent changes in local economic conditions, and support the interpretation that the identifying variation primarily reflects differences in broadband deployment costs.

Political Targeting of the Highway Network

The Interstate Highway System was initially designed to connect major cities without passing through them. However, the Bureau of Public Roads, in charge of designing and constructing the network, revised the original plan to secure Congressional approval. This resulted in the Yellow Book, a directory of additional routes passing through targeted urban areas. These modifications were often met with local opposition, as the new routes frequently displaced established neighborhoods (Murphy, 2009; Strömberg, 2016). If municipalities in Yellow Book target areas retain distinct political characteristics that affect federal transfers, proximity to these routes could violate the exclusion restriction through a channel unrelated to broadband deployment.

Table D.4 reports 2SLS estimates that include an indicator for municipalities affected by the Yellow Book revisions interacted with year fixed effects. This specification allows municipalities located along politically targeted highway routes to follow flexible, time-varying trends in federal spending. The results remain unchanged and, if anything, slightly larger in magnitude, indicating that the political targeting of highway placement does not drive the findings.

Table D.4: The Effect of Broadband Expansion on Federal Spending, controlling for Yellow-Book Municipalities

Dep. Var: Log(Spending per Capita)	Total	Mandatory	Discretionary	Loans & Insurance
	(1)	(2)	(3)	(4)
Number of ISPs	0.099*	0.033	0.157**	0.106**
	(0.054)	(0.077)	(0.061)	(0.052)
Kleibergen-Paap F-test	33.742	33.742	33.742	33.742
Avg per Capita Spending at Baseline	1,908	97	764	1,047
Municipalities	35,672	35,672	35,672	35,672
Observations	428,064	428,064	428,064	428,064

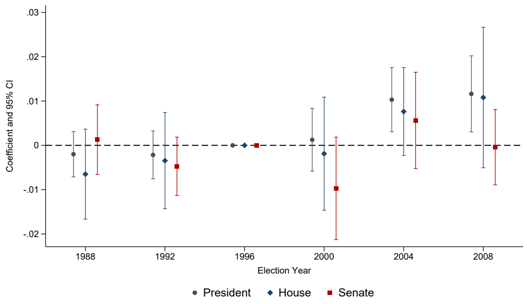
Notes: This table reports 2SLS estimates of the effect of Broadband expansion on municipality federal spending, controlling for Yellow-Book municipalities. The dependent variable is the logarithm of one plus federal spending per capita in municipality m and year t, and the number of ISPs in a municipality is instrumented with the interaction between municipality proximity to Interstate highways and a linear time trend. All specifications are weighted by municipality population in 2000. Robust standard errors clustered at the state level are reported in parentheses. The Kleibergen–Paap F-statistic is reported for the 2SLS specifications. Significance levels: *** 1%, ** 5%, * 10%.

Pre-existing Trends in Campaign Contributions

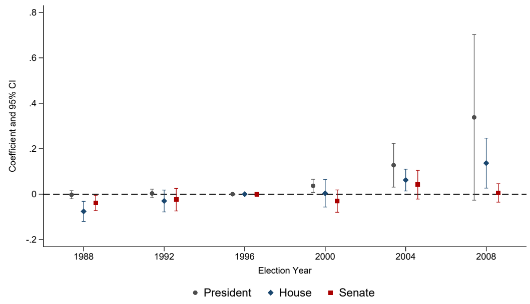
Figure D.4 reports semi-parametric reduced-form estimates of the effect of proximity to Interstate highways on campaign contributions across election cycles from 1988 to 2008. The sample covers presidential, House, and Senate elections, and the four panels correspond to the outcome variables: total amount contributed, number of contributions, first-time contributions, and small contributions. Coefficients are flat and statistically indistinguishable from zero prior to the 2000 election cycle, providing no evidence of differential pre-existing trends in political giving across municipalities with different levels of highway proximity. Contributions begin to diverge from the 2000 cycle onward, and the pattern is most pronounced for presidential and House elections, consistent with the results of Table 4.

Figure D.4: Reduced-Form Effects of Proximity on Campaign Contributions

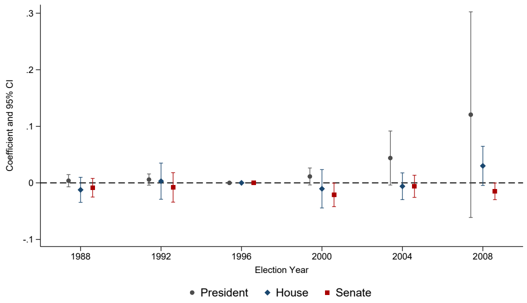
(a) Amount Contributed



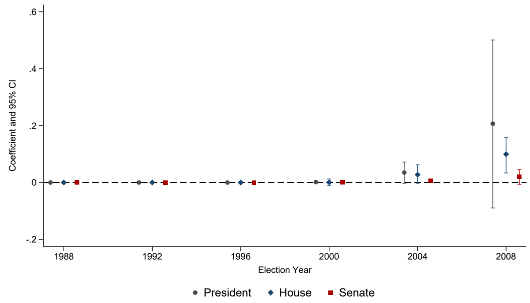
(b) Number of Contributions



(c) First-time Contributions



(d) Small Contributions



Notes: This figure reports semi-parametric reduced-form estimates of the effect of proximity to Interstate highways on campaign contributions using the preferred specification. The dependent variable in panel (a) is the logarithm of one plus total contributions per capita; in panel (b), the logarithm of one plus the number of contributions per capita; in panel (c), the logarithm of one plus the number of first-time contributor contributions per capita; and in panel (d), the logarithm of one plus the number of small contributions per capita, where small contributions are defined as donations below \$200. Circles denote presidential elections, diamonds House elections, and squares Senate elections. All regressions are weighted by municipality population in 2000. Robust standard errors clustered at the state level are used to construct 95% confidence intervals.